Lecture 22 Notes

Recap: Full and Reduced SVD

The instructor recaps the SVD established in L21.

- **Singular values:** the diagonal entries of Σ, real and non-negative; only r of them are nonzero.
- **Left singular vectors:** $u_1, \dots, u_m$ (columns of U); **right singular vectors:** $v_1, \dots, v_n$ (columns of V).
- **Full SVD:** $A = U\Sigma V^*$ with U $(m \times m)$ and V $(n \times n)$ **unitary** and Σ $(m \times n)$ rectangular diagonal.
- **Reduced SVD:** eliminate the zero diagonal entries of Σ (which silence the corresponding columns of U and V); what remains are the **non-degenerate principal semi-axis directions** $u_1, \dots, u_r$ and their pre-images $v_1, \dots, v_r$, giving $A = \hat{U}\hat{\Sigma}\hat{V}^*$ with an $r \times r$ positive diagonal.
- **Outer-product form:** $A = \sum_{k=1}^{r} \sigma_k u_k v_k^*$, a sum of rank-1 terms that (with respect to the matrix inner product) are **orthogonal**. The instructor describes this as writing A in a matrix basis **adapted to** A: the full family can be completed from singular-vector outer products, while A itself uses only the diagonal/rank-1 SVD terms weighted by σ_k.

The geometric origin: the image of a hypersphere under A is an **ellipsoid**; the σ_k are the principal semi-axis lengths, the u_k the axis directions, the v_k their pre-images.

SVD as the Ultimate Matrix-Analysis Tool

The instructor stresses the SVD is a **perfect matrix / linear-mapping analysis tool** — "like the miracle products they sell on TV," advertising feature after feature. Given the SVD of A, you can read off:

1. **Rank:** the number of nonzero singular values $= r$ (the number of non-degenerate ellipsoid dimensions = dimension of range = dimension of row space).
2. **Column space $\mathcal{R}(A)$:** orthonormal basis $u_1, \dots, u_r$ (first r columns of U).
3. **Left null space $\mathcal{N}(A^*)$:** $u_{r+1}, \dots, u_m$ (orthogonal to the column space).
4. **Row space $\mathcal{R}(A^*)$:** $v_1, \dots, v_r$ (first r columns of V).
5. **Null space $\mathcal{N}(A)$:** $v_{r+1}, \dots, v_n$.
6. **Induced 2-norm:** $\|A\|_{2,2} = \sigma_1$ (recall the proof started by setting $\sigma_1 = \|A\|_{2,2}$).
7. **Frobenius norm:** $\|A\|_F = \sqrt{\sum_k \sigma_k^2}$ (derived below).

Projecting a vector onto the range of A is then **easy**: take inner products with the orthonormal $u_1, \dots, u_r$ and recombine.

Why Singular Values Are Real and Non-Negative

Geometric reason. In the construction, $\sigma_1 = \|A\|_{2,2}$ is a **norm** (a length of a principal semi-axis), hence real and ≥ 0. Each subsequent σ_k is the 2-norm of a smaller block in the recursive proof — also a norm. So all σ_k are non-negative reals; off-diagonal entries of Σ are zero. Hence Σ is a **real non-negative diagonal** matrix, and the count of nonzero diagonal entries is the rank.

Algebraic reason (via A^*A). From $A = U\Sigma V^*$,

$$A^*A = V\Sigma^*U^*U\Sigma V^* = V(\Sigma^*\Sigma)V^*,$$

the eigendecomposition of A^*A with eigenvalues σ_k^2 (diagonal of $\Sigma^*\Sigma$) and eigenvectors V. Since $A^*A \succeq 0$ (proved earlier: $x^*A^*Ax = \|Ax\|^2 \geq 0$), the eigenvalues $\sigma_k^2 \geq 0$, so $\sigma_k = \sqrt{\lambda_k(A^*A)}$ is a real non-negative number.

Frobenius Norm from Singular Values

$$\|A\|_F = \sqrt{\operatorname{tr}(A^*A)}.$$

Substitute the SVD. With $A = U\Sigma V^*$ and $A^* = V\Sigma^\top U^*$ (order reverses; Σ is real):

$$\operatorname{tr}(A^*A) = \operatorname{tr}(V\Sigma^\top U^*U\Sigma V^*) = \operatorname{tr}(V\Sigma^\top\Sigma V^*).$$

The $U^*U = I$ cancels; then by the cyclic property $\operatorname{tr}(VXV^*) = \operatorname{tr}(V^*VX) = \operatorname{tr}(X)$:

$$\operatorname{tr}(A^*A) = \operatorname{tr}(\Sigma^\top\Sigma).$$

Now $\Sigma^\top\Sigma$ is diagonal with entries $\sigma_1^2, \dots, \sigma_r^2, 0, \dots$, whose trace is $\sum_k \sigma_k^2$. Therefore

$$\boxed{\|A\|_F = \sqrt{\sigma_1^2 + \cdots + \sigma_r^2}}.$$

Alternative (unitary invariance). Frobenius is unitarily invariant, so $\|A\|_F = \|U\Sigma V^*\|_F = \|\Sigma\|_F = \sqrt{\sum_k \sigma_k^2}$ directly. The instructor phrases this as the **energy of the singular-value vector**. Both $\|A\|_{2,2} = \sigma_1$ and $\|A\|_F$ depend **only on the singular values**, because both are unitarily invariant.

Computing the SVD via A^*A or AA^*

This is **not** the numerically preferred algorithm, but it shows the **connection between SVD and eigendecomposition**.

- **Via A^*A:** $A^*A = V(\Sigma^\top\Sigma)V^*$. It is Hermitian and **positive semidefinite** (always), so it has a unitary eigendecomposition. Its eigenvectors are the **right singular vectors** V; its eigenvalues are σ_k^2, giving the singular values. Then recover the left singular vectors from $Av_k = \sigma_k u_k$, i.e. $u_k = Av_k/\sigma_k$ — **no second eigendecomposition needed**.
- **Via AA^*:** $AA^* = U(\Sigma\Sigma^\top)U^*$; its eigenvectors are the **left singular vectors** U.

(Shape note: if A is fat, A^* is tall; if A is full-rank fat then AA^* is positive **definite**, but A^*A or AA^* is always at least positive **semidefinite** and Hermitian — hence normal, hence unitarily diagonalizable.)

Numerical warning. In practice forming A^*A is **inadvisable** (it squares the condition number). Built-in routines (MATLAB `svd`, NumPy `numpy.linalg.svd`) work directly on A (bidiagonalization), avoiding A^*A. The instructor doesn't know the exact state-of-the-art algorithm but confirms it avoids the explicit product.

Low-Rank Approximation (Eckart–Young)

A major application area.

Problem. Given A, find a rank-p matrix B as close to A as possible in Frobenius norm:

$$\min_{\operatorname{rank}(B)\le p} \|A - B\|_F.$$

The transcript states the constraint as "rank p"; the standard formulation is rank **at most** p. The truncated SVD has rank at most p, and it has rank exactly p when the first p retained singular values are nonzero.

Why it's hard. The objective (a norm) is **convex** — "a pleasure to optimize." But the **constraint set** (rank-p matrices) is **not convex**: the convex combination of two rank-p matrices can have rank up to $2p$. (Example: two orthogonal rank-1 matrices averaged 0.5/0.5 give a rank-2 matrix.) So the feasible region is non-convex, making the problem hard in general.

Solution (Eckart–Young). Despite non-convexity, the SVD gives the **exact global optimum**. Because the singular values are **ordered** $\sigma_1 \geq \sigma_2 \geq \cdots$ (monotonically non-increasing, by construction), simply **truncate** the outer-product sum at p:

$$A_p = \sum_{k=1}^{p} \sigma_k u_k v_k^*,$$

discarding the $r - p$ smallest singular values. The error is

$$\|A - A_p\|_F = \sqrt{\sum_{k=p+1}^{r} \sigma_k^2}.$$

The instructor does **not** prove this theorem in lecture; he states the result and moves to implications/applications.

The same truncation is optimal in the induced 2-norm, with error $\|A - A_p\|_{2,2} = \sigma_{p+1}$. The instructor notes he mentions this only after emphasizing the ordering, which is "critical" for this application. For other norms, e.g. the **1-1 norm**, he says he does not know a closed-form solution; changing the norm can force non-convex heuristics. Thus the **choice of norm determines both the meaning of "close" and the tractability** of the problem.

Application: Image Compression

A grayscale image is a matrix; the running example is the **guitar-player** photo. The source aside identifies the musician with Deep Purple, but the transcribed name is garbled. The image began as color/RGB and was converted to a monochrome intensity matrix of size $1236 \times 2060 \approx 2.5$ million pixel values.

Compute the SVD and keep only the first k rank-1 terms: $A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^*$. **Storage:** instead of mn numbers, store k singular values, km numbers for the u's, and kn for the v's — total $k(m + n + 1)$. For $m \approx 1000, n \approx 2000$: - **Rank-1** ($k = 1$): you "barely have some idea" what the image is — a poor representation. - **Rank-10:** $\approx 10 \cdot 1000 + 10 \cdot 2000 + 10 \approx 30{,}000$ numbers vs. 2 million. - **Rank-20:** $\approx 60{,}000$ vs. 2 million.

The point of the storage count is that we store the **factors** u_i, v_i, and σ_i, not the already-multiplied reconstructed image. The instructor repeatedly frames the comparison as roughly **2 million full pixel values** versus **30,000** (rank 10) or **60,000** (rank 20) stored factor entries.

The error (in Frobenius/MSE sense, since Frobenius measures squared error) is small for modest k because the **singular values decay rapidly** for natural images (the first is large; most are near zero — best seen on a log scale). This is **not** how real image compression is done, but it illustrates the principle: represent the data in a basis where energy concentrates in a few coefficients. (Netflix matrix completion — discussed later — is another low-rank application.)

Closest Unitary Matrix to a Given Matrix

A second SVD application. **Problem:** given a square matrix A (not unitary), find the unitary Q ($QQ^* = I$) closest to it in **Frobenius** norm:

$$\min_{QQ^*=I} \|A - Q\|_F^2.$$

Again the objective is convex but the **set of unitary matrices is non-convex** (the convex combination of two unitaries need not be unitary), so it looks hard. But the SVD gives a **closed-form** solution.

The Frobenius norm is chosen deliberately: the instructor says the problem is hard for other norms, while Frobenius gives a clean analytic formula. He also mentions that unitary approximation appears in some recent neural-network algorithms.

Scalar analogy (1×1). For a complex number $a = re^{j\theta}$, the closest unit-magnitude number is $e^{j\theta}$ — obtained by the **polar decomposition** and replacing r with 1.

Derivation. Write $A = U\Sigma V^*$ and use Frobenius unitary invariance:

$$\|A - Q\|_F = \|U\Sigma V^* - Q\|_F = \|\Sigma - U^*QV\|_F = \|\Sigma - Q'\|_F,$$

where $Q' = U^*QV$ is unitary (product of unitaries). Minimizing the distance from a **non-negative diagonal** Σ to a unitary Q' is solved by $Q' = I$. Hence

$$\boxed{Q = UV^*},$$

i.e. **find the SVD, set all singular values to 1, and multiply** UV^* — clearly unitary, and the closest unitary to A in Frobenius norm.

Application: Rigid Motion / Image Registration (Computer Vision)

Setup. In one scene a (rigid) object sits at some location; in a second scene it has **moved without changing shape**. We want the motion = **rotation** (a unitary/orthogonal Θ) **plus translation** t. Register matching points $x_1, x_2, \dots$ in the first scene and $y_1, y_2, \dots$ in the second; ideally each $y_i = \Theta x_i + t$. Stacking points as columns of X and Y (and adding t to each column via $t\,\mathbf{1}^\top$):

$$Y \approx \Theta X + t\,\mathbf{1}^\top.$$

Why it's not exact: registration noise, and projecting a 3-D world to 2-D. So minimize the **least-squares** error:

$$\min_{\Theta,t} \left\| Y - \Theta X - t\,\mathbf{1}^\top \right\|_F^2,$$

which (as a least-squares problem, revisited under inner-product spaces) has a nice solution — **but** it ignores the **constraint** that Θ be unitary ($\Theta\Theta^* = I$). Two strategies: 1. **Project after solving:** solve the unconstrained least squares to get a general $\Theta^\star$, then find the **closest unitary matrix** to $\Theta^\star$ via UV^*. (Simple but not optimal.) 2. **Gradient on the unitary manifold:** take a gradient step that leaves the unitary set, then **project back** onto the unitary set (via UV^* from the SVD), repeating — moving "a little bit and coming back" along the manifold.

Related ideas appear in **independent component analysis (ICA)** (minimize mutual information over a unitary matrix, after a PCA whitening step). These are mentioned as buzzwords; details deferred.

The projected-gradient picture is not presented as the only or best constrained method. The instructor says there are many algorithms; in practice one may use something other than the raw gradient, and he skips those details.

Polar Decomposition for Matrices

The complex polar form $c = re^{j\theta}$ ($r > 0$, $e^{j\theta}$ on the unit circle) generalizes to matrices. The instructor calls this a **non-trivial decomposition**: a **non-singular square** A can be written

$$A = PT, \qquad P \succ 0 \text{ (positive definite)}, \quad T \text{ unitary}.$$

("P" plays the role of $r > 0$; T the role of $e^{j\theta}$.) **Why it's easy from the SVD:** for non-singular A, Σ is positive (full rank). Insert $U^*U = I$:

$$A = U\Sigma V^* = (U\Sigma U^*)(UV^*).$$

Here $U\Sigma U^*$ is **positive definite** (unitary diagonalization with positive diagonal), and UV^* is **unitary**. So $P = U\Sigma U^*$, $T = UV^*$. **Notably, the unitary factor $T = UV^*$ is exactly the closest unitary matrix to** A in Frobenius norm (from the previous section).

1×1 **check and the correspondence.** In the scalar case P is a positive real number ($= r$) and T is a number on the unit circle ($= e^{j\theta}$) — recovering the complex polar form. A **student asks** about a real/imaginary-part decomposition; the instructor clarifies the right correspondence is via **eigenvalues**, not real/imaginary parts: a general complex matrix lives in an n^2-dimensional space, so you cannot split it into "two dimensions." The clean analogy is: - **Hermitian** matrices $\leftrightarrow$ **real** numbers (eigenvalues on the real line), - **positive definite** matrices $\leftrightarrow$ **positive real** numbers, - **unitary** matrices $\leftrightarrow$ numbers on the **unit circle** (eigenvalues on the unit circle). So a unitary matrix generalizes a point on the unit circle, and the polar decomposition generalizes $c = re^{j\theta}$.

Schatten p-Norms

New norms defined from the **singular values**. Collect the singular values into a vector $\sigma = (\sigma_1, \dots, \sigma_r)$. The **Schatten p-norm** of A is the **vector p-norm of** σ:

$$\|A\|_{(p)} = \Big(\sum_k \sigma_k^p\Big)^{1/p}.$$

(The subscript (p) is written on the **left/inside** to distinguish it from the induced norm.) Special cases: - $p = 2$: $\|A\|_{(2)} = \sqrt{\sum_k \sigma_k^2} = \|A\|_F$ — the **Frobenius** norm. - $p = \infty$: $\|A\|_{(\infty)} = \sigma_1 = \|A\|_{2,2}$ — the **operator** norm. - $p = 1$: $\|A\|_{(1)} = \sum_k \sigma_k$ — the **nuclear norm** (below).

So Frobenius and operator norms are both Schatten norms; the new one is $p = 1$.

Nuclear Norm and Low-Rank via Convex Relaxation

The **Schatten-1 norm** — the **sum of singular values** — is "the star of the show," researched intensively for two decades. It has its own name and notation:

$$\|A\|_* = \sum_k \sigma_k \quad \text{(nuclear norm, a.k.a. trace norm)},$$

sometimes written as the trace of $(A^*A)^{1/2}$.

Why it matters — convex relaxation of rank. Recall ℓ_1 minimization promotes **sparsity** (many zero entries). Applying the same idea to the **singular value vector** σ: minimizing the nuclear norm $\|A\|_*$ drives most σ_k to **zero** — and the number of nonzero singular values is the **rank**. So **minimizing the nuclear norm minimizes the rank** (with appropriate constraints on A; otherwise the trivial minimizer is the zero matrix). The nuclear norm is the **convex substitute for rank**, just as ℓ_1 is the convex substitute for sparsity. This gives **low-rank solutions** through a tractable convex problem.

Warning / exam-style caution. The instructor explicitly flags the same caution as in ℓ_1/CVX-style sparsity problems: minimizing the nuclear norm alone gives the zero matrix. It becomes meaningful only with constraints, e.g. matching observed entries in a completion problem.

Application: Matrix Completion and the Netflix Challenge

Who wants low rank? Matrix completion — central to **recommendation systems** and the **Netflix challenge**.

Setup. Rows = customers (tens/hundreds of millions), columns = movies (perhaps hundreds of thousands). Entry (i, j) is customer i's rating of movie j, but each customer has rated only a few movies, so the matrix is **mostly incomplete**. Goal: **fill in** the missing entries to predict whether a customer would like an unseen movie.

Low-rank model. Represent the rating matrix as a product of two thin matrices:

$$\text{Ratings} \approx P\,Q^\top,$$

where each customer is an r-dimensional **preference vector** (a row of P) and each movie is an r-dimensional **feature vector** (a row of Q). The **inner product** of a user-preference vector and a movie-feature vector gives the predicted **score**. The features may correspond to interpretable factors (is it a horror movie? a comedy? a certain actor present?); a user who weights "comedy" highly scores a comedy highly.

Two approaches: 1. **Fixed-rank factorization:** choose rank r (e.g. 100), and minimize the **Frobenius error over the known entries**:

$$\min_{P,Q} \;\big\| \text{Ratings} - PQ^\top \big\|_F^2 \quad \text{(over observed entries)},$$

recovering the customer matrix P and movie matrix Q. This **low-rank decomposition is the approach that won the Netflix challenge.** 2. **Nuclear-norm minimization:** minimize $\|X\|_*$ subject to matching the observed entries — a convex relaxation that produces a low-rank completion without fixing r in advance.

History: the instructor describes the Netflix challenge as an early-2000s event (the transcript says "2003 / beginning of 2000") where Netflix released ratings data but held out some values; teams who best predicted the hidden values won a **$1,000,000 prize**, and the winning solutions used such low-rank approximation.

The next topic is **inner product spaces**; the instructor says he will continue this discussion on Thursday. He also ends with an administrative request for students to complete a class/course item (the transcript wording is garbled).

Instructor Remarks and Study Guidance

- The SVD is the **ultimate analysis tool**: rank, orthonormal bases for all four fundamental subspaces, $\|A\|_2 = \sigma_1$, and $\|A\|_F = \sqrt{\sum \sigma_k^2}$ — all read off directly.
- Singular values are **real, non-negative** because they are norms (geometric) and because $\sigma_k^2 = \lambda_k(A^*A) \geq 0$ with $A^*A \succeq 0$ (algebraic).
- $\|A\|_F = \sqrt{\sum \sigma_k^2}$ via the trace cyclic property or unitary invariance.
- **Eckart–Young:** truncating the ordered SVD gives the best rank-p approximation (global optimum despite the non-convex rank constraint), in both Frobenius and 2-norms; image compression is the showcase.
- **Closest unitary matrix** to A (Frobenius) is $Q = UV^*$ (set all $\sigma_k = 1$); used in rigid-motion/registration and ICA.
- **Polar decomposition** $A = (U\Sigma U^*)(UV^*) = PT$ generalizes $c = re^{j\theta}$; the correspondence is via eigenvalues (Hermitian↔real, PD↔positive real, unitary↔unit circle).
- **Schatten p-norms** are p-norms of the singular-value vector: $p = 2 \to$ Frobenius, $p = \infty \to$ operator, $p = 1 \to$ **nuclear norm** ($\sum \sigma_k$), the **convex relaxation of rank** (low-rank analog of ℓ_1 sparsity).
- **Matrix completion / Netflix:** model ratings as low-rank $PQ^\top$; the winning approach minimized Frobenius error over observed entries; nuclear-norm minimization is the convex alternative.
- **Recurring warnings:** changing the norm can destroy the closed-form SVD solution; non-convex constraints appear in both rank-p and unitary-matrix problems; nuclear-norm minimization needs additional constraints or the zero matrix wins.
- **Closing:** continuation on Thursday, then inner product spaces; administrative class/course request at the end.

Source and Coverage Note

Source: `corrected/lecture22_corrected.md`.

Coverage: Recap of full/reduced SVD and outer-product form; SVD as analysis tool (rank, four fundamental subspaces, projection to range, σ_1, Frobenius); why singular values are real/non-negative (geometric and algebraic via $A^*A \succeq 0$); Frobenius norm $= \sqrt{\sum \sigma_k^2}$ (trace-cyclic and unitary-invariance derivations, singular-value energy); computing the SVD via A^*A/AA^* eigendecomposition (both directions, $u_k = Av_k/\sigma_k$, PSD/Hermitian/normal structure, numerical warning); low-rank approximation / Eckart-Young (rank p versus rank at most p, convex objective, non-convex rank set with the rank-1+rank-1 example, ordered-σ truncation solution stated without proof, Frobenius and 2-norm errors, no closed form for other norms); image compression (guitar-player/Deep Purple aside with garbled name, RGB-to-monochrome conversion, 1236×2060, storage $k(m + n + 1)$, factor storage rather than full product, rank-1/10/20 comparison, singular-value decay, not real image compression); closest unitary matrix ($Q = UV^*$ via Frobenius invariance, deliberate Frobenius choice, scalar polar analogy, neural-network mention); rigid-motion/image-registration application (rotation+translation, least squares, ignored unitary constraint, project-onto-unitary vs. projected-gradient/manifold method, many algorithms/raw-gradient caveat, ICA/PCA mention); matrix polar decomposition ($A = (U\Sigma U^*)(UV^*)$, nonsingular square assumption, non-triviality, unitary factor = closest unitary, scalar check, eigenvalue correspondence Hermitian/PD/unitary, student Q&A on real/imaginary parts); Schatten p-norms (p-norm of singular-value vector; notation distinction; $p = 2$ Frobenius, $p = \infty$ operator, $p = 1$ nuclear); nuclear/trace norm as convex relaxation of rank (sparsifying singular values, zero-solution constraint warning); matrix completion and the Netflix challenge (incomplete ratings matrix, low-rank $PQ^\top$ user/movie factorization, fixed-rank Frobenius vs. nuclear-norm approaches, $1M prize and source-stated early-2000s history); closing continuation on Thursday, inner product spaces, and garbled administrative request.